YISHUN INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
Higher 1

NAME

INDEX NO | CG

BIOLOGY — 8876/02

Paper 2 Structured Questions and Free Response Questions

1 Sep 2021

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name, index no. and CG on this cover page.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions in the spaces provided on the Question paper.

Section B
Answer any **one** question in the spaces provided on the Question Paper. Indicate the question you have attempted at the top of page **18**.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.
At the end of the examination, fasten all your work securely together.

For Examiner's Use	
Section A	
1	7
2	15
3	13
4	10
Section B	15
Total	60

Overall Results for JC2 H1 Biology Preliminary Examination				
Paper	Raw Marks	Weighting	%	Grade
1	30			
2	60			

This document consists of **19** printed pages and **1** blank page.

Section A

Answer **all** questions in this section.

1 To investigate the amount of time cells spent in each phase of the cell cycle, a population of dividing diploid cells is stained with a dye that becomes fluorescent when it binds to DNA. The amount of fluorescence is directly proportional to the amount of DNA in each diploid cell.

To measure the amount of DNA in each cell, the population of dividing diploid cells are passed through a fluorescence-activated cell sorter (FACS), an instrument that measures the amount of fluorescence in individual cells. The number of cells with a given DNA content, in each phase of the cell cycle, is plotted on a graph shown in Fig. 1.1.

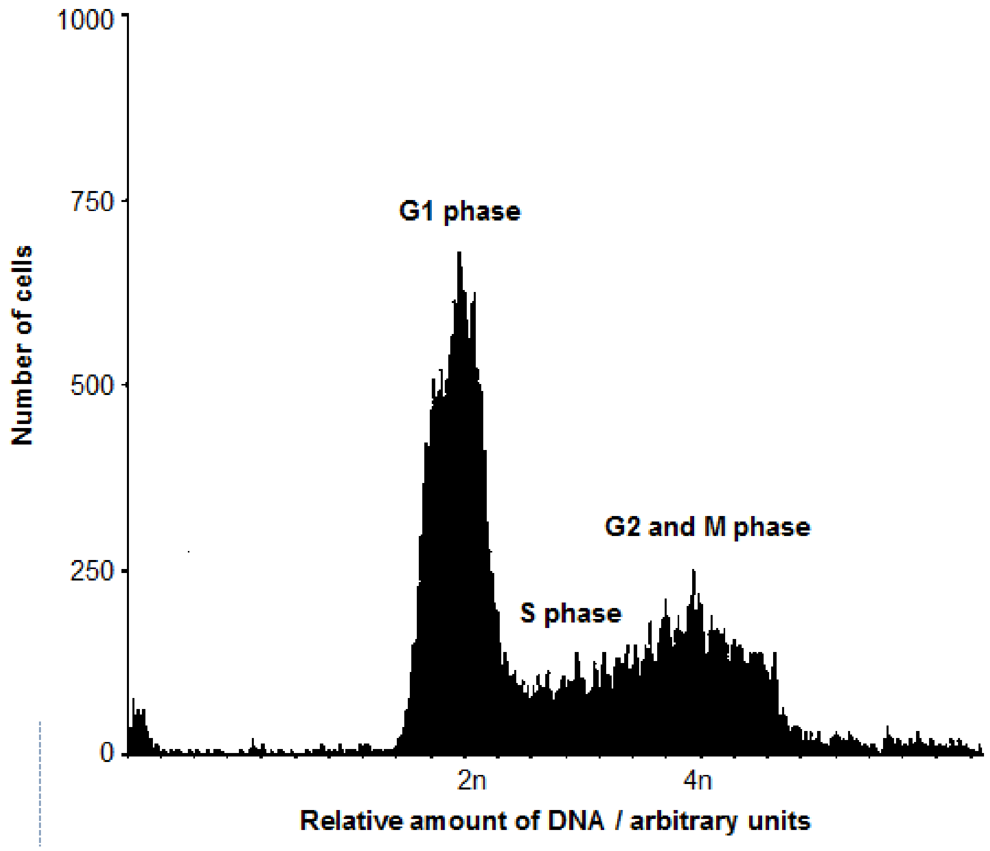

Fig. 1.1

(a) With reference to Fig. 1.1,

(i) explain how the relative amount of DNA changes as the cells progress through the different phases of the cell cycle.

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.... [3]

(ii) suggest why the G1 phase is the longest phase of the cell cycle in this population of cells.

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.... [1]

(b) Cell division is *almost* identical in animal and plant cells.

Describe three differences in cell division between an animal cell and a plant cell.

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.... [3]

[Total: 7]

2 A key reaction in photosynthesis occurs when ribulose bisphosphate carboxylase (Rubisco) catalyses the fixation of carbon dioxide to ribulose bisphosphate (RuBP).

Fig. 2.1 shows the summary of events and stages occurring during the translation of Rubisco mature mRNA.

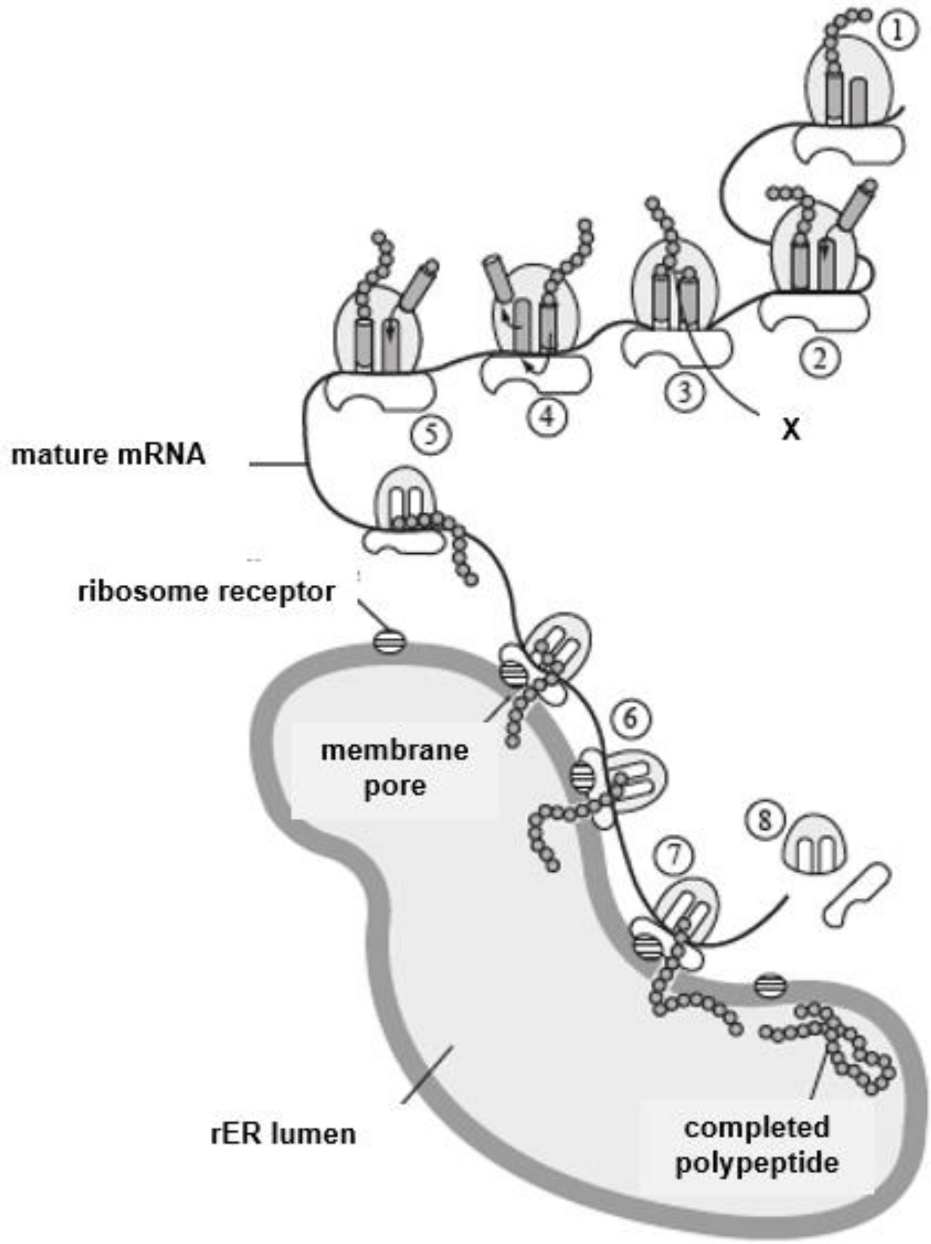

Fig. 2.1

(a) With reference to Fig. 2.1,

(i) name the bond labelled **X**.

.... [1]

(ii) outline the processes occurring in stages **4** and **5**.

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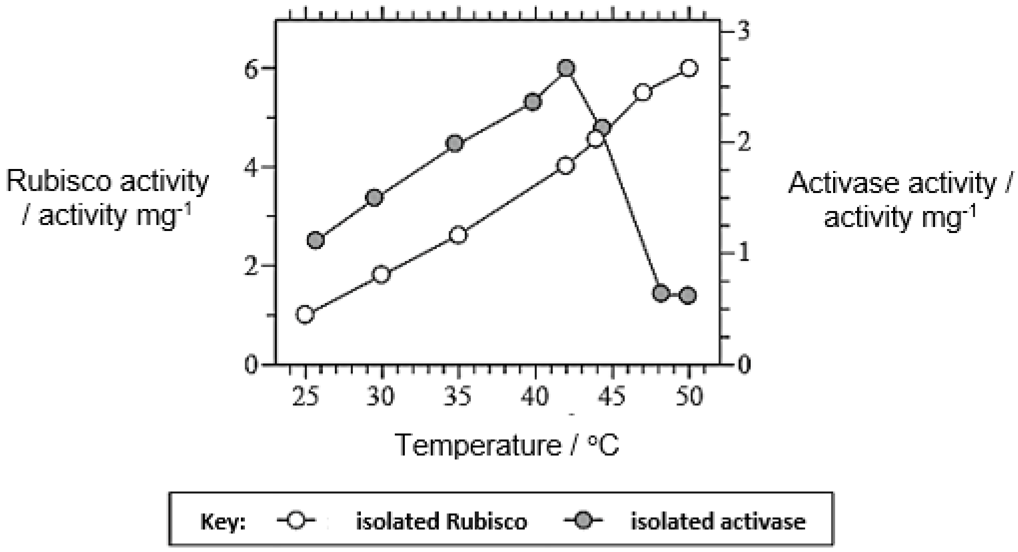
.... [2]

To be effective, Rubisco must be activated by another enzyme called activase. The activities of Rubisco and activase, each isolated from tobacco leaves, were independently investigated in a laboratory, under conditions of increasing temperature.

Fig. 2.2 shows the effect of temperature on the activities of the two enzymes.



[Source: adapted from S. Crafts-Brandner and M. Salvucci (2000) 'Rubisco activase constrains the photosynthetic potential of leaves at high temperature and CO2.' *PNAS*, 97, pp. 13 430–13 435. Figure 2.]

Fig. 2.2.

(b) With reference to Fig. 2.2,

(i) describe the relationship between Rubisco activity and temperature.

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.... [1]

(ii) explain the changes in activase activity at temperatures higher than 42°C.

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.... [3]

(c) In a leaf, both enzymes are present together. Predict, with a reason, how the rate of photosynthesis would change from 35°C to 50°C.

[2]

(d) Light is not a factor that directly limits the rate of the Calvin cycle but it is needed for the Calvin cycle to continue.

Outline why light is needed for the Calvin cycle to continue.

[2]

In plants, the carbohydrates manufactured in photosynthesis are used to form polysaccharides such as starch.

(e) Explain how the structure of starch is adapted for its function in plants.

[4]

[Total: 15]

3 There have been many breakthroughs in stem cell research in recent years. It has been discovered that stem cells are involved in the replacement of worn-out cells and repair of damaged tissues. Further research is being conducted to better understand the mechanism involved in controlling the behaviour of stem cells in order to better manipulate them to treat various diseases and disorders.

(a) **(i)** State the type of stem cells involved in the replacement of worn-out cells and repair of damaged tissues

[1]

(ii) Describe **two** unique properties of this type of stem cells.

[2]

(b) Explain the importance of mitosis in the repair of damaged tissue.

[1]

(c) Induced pluripotent stem cells (iPSCs) are promising tools for cell replacement therapy. This is a medical procedure in which living iPSCs are injected into a patient to treat certain genetic diseases.

State **one** advantage of using iPSCs for cell replacement therapy.

[1]

(d) Scientists have detected telomerase in more than 90% of human tumour samples. Stem cells also express high levels of telomerase and can divide indefinitely.

Explain why normal adult stem cells do not give rise to tumour, despite having a high level of telomerase activity.

[2]

The appearance of cancer cells has been known to be different from normal cells. These differences have been used as a method of diagnosis by doctors. Prostate cells taken from wild-type mice and mice with prostate cancer were analysed.

Fig. 3.1 compares the same organelle found in these cells viewed under the electron microscope, while Fig. 3.2 shows the levels of ribosomal RNA (rRNA) measured in these two types of cells.

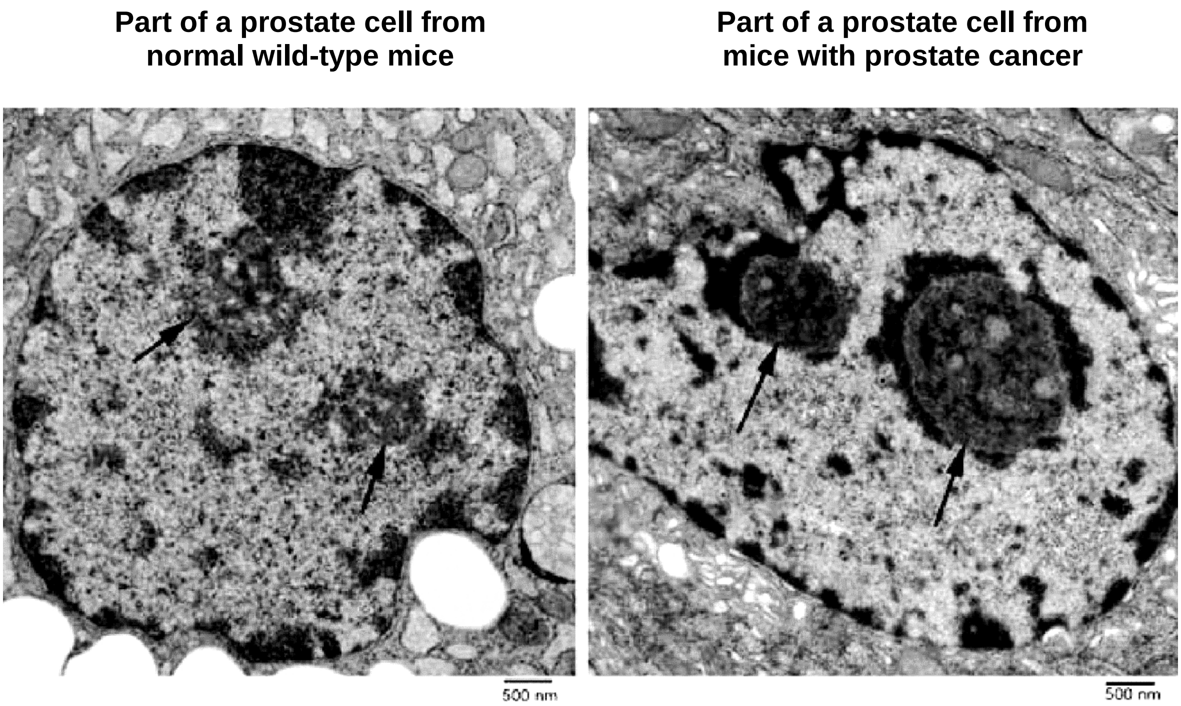


Fig. 3.1

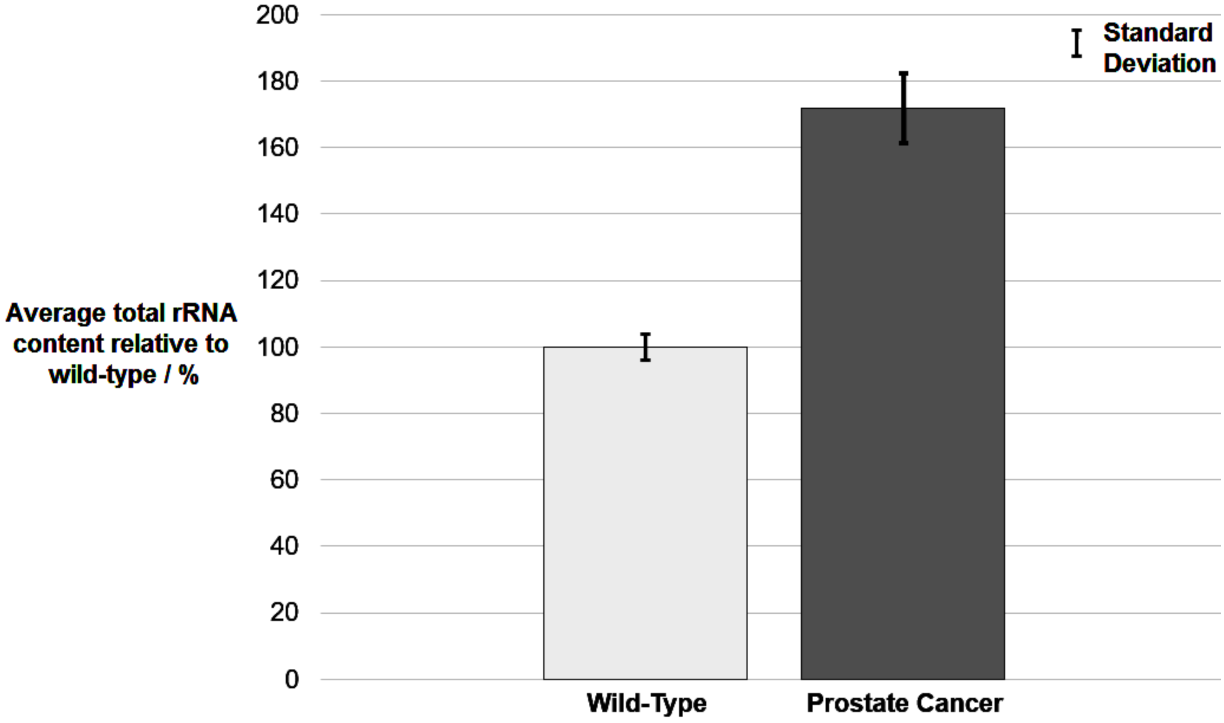


Fig. 3.2

(e) **(i)** Identify the regions indicated by the arrows in Fig. 3.1.

[1]

(i) With reference to the regions identified in Fig. 3.1 and information given in Fig. 3.2, describe the differences between normal prostate cells and prostate cancer cells.

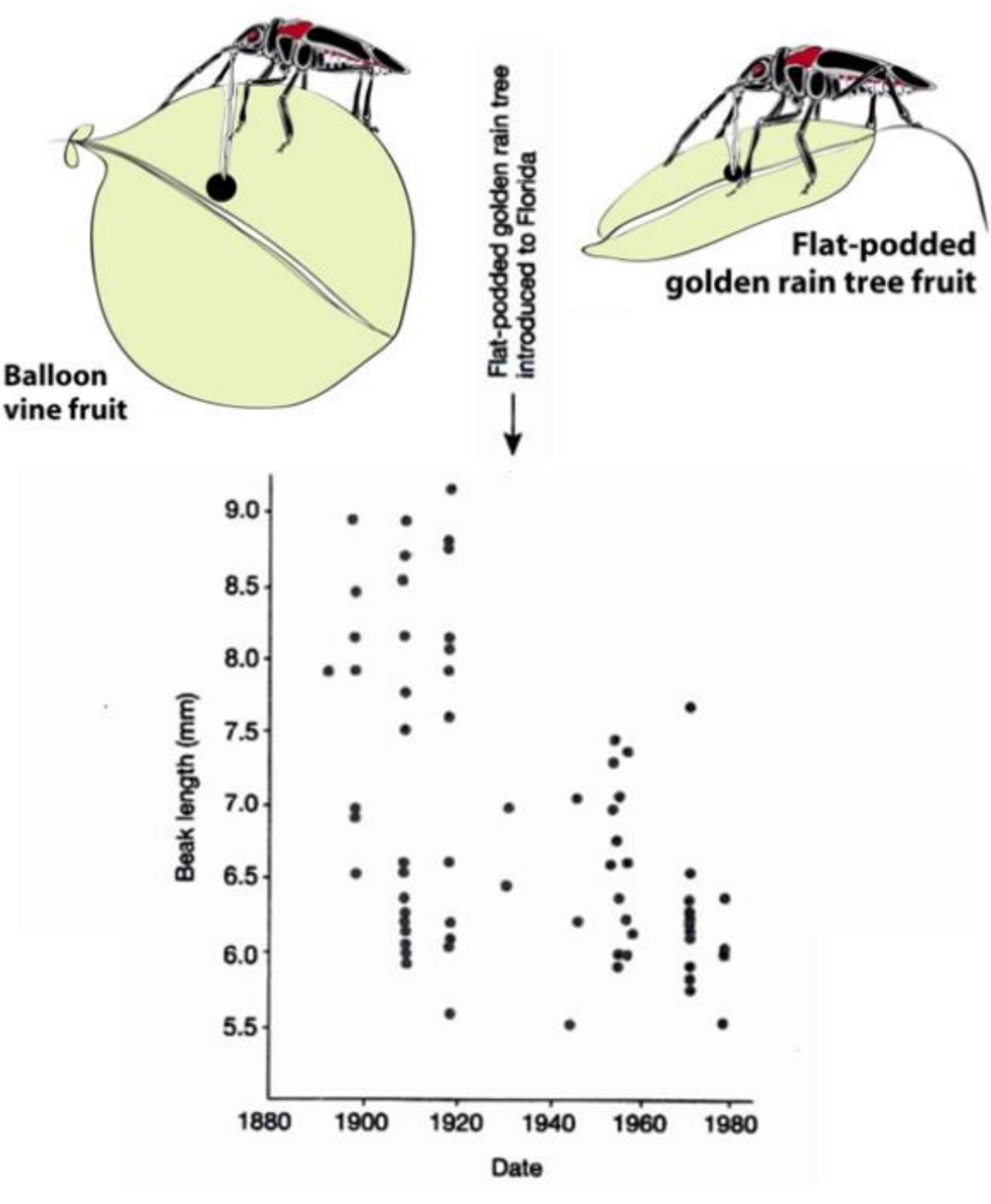
[2]

(iii) Explain the differences described in **(e)(ii).**

[3]

[Total: 13]

4 Soapberry bugs in Florida originally fed on the native balloon vine, using their sharp beaks to penetrate the fruits to reach the seeds. After the introduction of the flat-podded golden rain tree from Asia to mainland central Florida, the phenotype of the soapberry bug has changed, as shown in Fig. 4.1.



Legend: Each dot on the graph shows the average beak length of 1000 bugs of a population.

Fig. 4.1

(a) With reference to Fig. 4.1, describe how the phenotype of the soapberry bugs has changed after the introduction of the flat-podded golden rain tree fruit in 1930.

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.... [1]

(b) The flat-podded golden rain tree fruit has thinner-skinned fruit compared to that of balloon vine fruit.

Explain how this has resulted in the change in soapberry bug's phenotype after 1930.

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.... [4]

(c) Describe **one** source of genetic variation accounting for the differences in beak lengths in the soapberry bugs.

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.... [1]

(d) In soapberry bugs, a recessive mutant allele on chromosome 2 leads to vestigial wings as opposed to the normal wing size.

A second eyeless gene on the X chromosome encodes for a transcription factor required for normal eye development. A recessive mutant allele at this gene locus leads to absence of eye structures as opposed to normal eyes.

A female fly heterozygous at both gene loci was crossed with a normal eyed male fly heterozygous for wing size.

Using the following symbols,

X^E:	dominant allele on X chromosome for normal eyes	**X^e:**	Recessive allele on X chromosome for eyeless
A:	dominant allele for normal wing size	**a:**	recessive allele for vestigial wings

draw a genetic diagram to show the possible offspring of the cross.

[4]

[Total: 10]

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

5 **(a)** Describe how a gene mutation can lead to a different protein product. [6]

(b) The expression of genes gives rise to proteins that affect the biochemical reactions and other processes in organisms.

Discuss, with examples, the importance of specific shapes of proteins in organisms. [9]

6 **(a)** Describe the role of membranes in photosynthesis and cellular respiration. [6]

(b) Discuss the suitability of DNA as a hereditary molecule and its significance in DNA replication. [9]

Question ____________

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